VIT

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:

Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	B.Tech CSE(BRS, BPS, BAI, BDS)	Semester	:	FALL 2025-26
Course Code & Course Title	:	BCSE355L - Cloud Architecture Design	Class Number	:	CH2025260101669 CH2025260101675 CH2025260101477 CH2025260101672 CH2025260101474 CH2025260102049 CH2025260102016 CH2025260102019 CH2025260102055 CH2025260102010 CH2025260101353
Faculty	:	Dr.LINDA JOSEPH Dr.SANKARI M Dr.MALINI A Dr.SUGUNA M Dr.RAJESH KUMAR Dr.POORNIMA S Dr.S VIGNESHWARI Dr.PRADEEP K Dr.TAMILARASI K Dr.SELLAM V Dr.KRITHIGA R	Slot	:	B2+TB2
tion	:	1 hour 30 Minutes	Max. Mark		50

eral Instructions: < Use this space to provide additional information such as graph sheet, data book etc.>

Write only your registration number on the question paper in the box provided and do not write other information

- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		ABC works as a cloud engineer for a company that manages multiple AWS accounts for development, testing, and production environments. Recently, several contractors have joined the team to work on a short-term project. The contractors must have access to launch and manage EC2 instances only in the development account, and only during business hours (9 AM–6 PM). They should not be able to delete resources or view billing information. The security team also requires that multi-factor authentication (MFA) be enforced for all contractor logins, and that all access be logged for auditing purposes. (a) Design an IAM solution for ABC's company to meet these requirements. Explain how you would use IAM Users, Groups, Policies,	10	CO2	K3

		and Roles to provide the required permissions, while following the principle of least privilege and meeting the security and compliance needs described. [5 Marks] (b) Explain how a multinational company should design its AWS infrastructure using Regions and Availability Zones to achieve low latency, comply with data residency laws, and ensure high availability with disaster recovery. [5 Marks]			
2		a) State three advantages of using different AWS storage services such as Amazon S3, Amazon EBS, and Amazon Glacier in a single architecture. [4 Marks] b) An e-learning platform needs to store large volumes of course videos, deliver them to students across different regions with minimal latency, and keep older course materials in long-term archival storage for compliance and reference. Recommend and justify the most suitable AWS storage solution for this requirement, considering scalability, durability, cost, and access performance, and explain your choice of object-based, block-based, or archival storage or a combination. [6 Marks]	10	CO4	K2
3		A gaming analytics company processes millions of player interaction records daily. The application backend runs on Amazon EC2, with separate environments for development, testing, and production. Workloads fluctuate significantly during peak gaming seasons. a) Recommend the most suitable EC2 instance types for compute-intensive analytics, balanced general workloads, and memory-optimized data processing. Justify your selections. [4 Marks] b) Propose an EC2 pricing strategy that balances cost and performance for the three environments, explaining when to use On-Demand, Reserved, or Spot Instances. [3 Marks] c) Describe how Auto Scaling and Elastic Load Balancing can be used together to maintain performance during traffic surges while optimizing costs. [3 Marks]	10	CO4	K4
4		A global media company, StreamWorld, delivers high-definition live sports streams and on-demand content to millions of viewers. The company hosts its applications on AWS and wants to ensure fast content delivery to international users, secure communication between multiple VPCs across different AWS regions, and seamless integration with its on-premises video editing facility.. a) Design and explain an AWS architecture diagram showing how multiple VPCs across regions can be interconnected for secure and efficient data sharing, including appropriate subnet organization, route tables, and security groups. [4 Marks] b) Explain how Amazon CloudFront can be used to optimize both live streaming and on-demand video delivery for global audiences, highlighting how it handles static and dynamic content while enhancing security. [3 Marks] c) Recommend an AWS networking service that can connect the AWS environment with the on-premises video editing facility, explaining its performance, security, and cost considerations. [3 Marks]	10	CO1	K3